STATISTICAL FACTSHEET 2015

PROVISIONAL VALUES AS OF 4 MAY 2016

European Network of
Transmission System Operators
for Electricity

entsoe
Reliable Sustainable Connected

ENTSO-E in figures – Electricity system data of member TSOs' countries

ENTSO-E Transmission network losses percentage consumption: 1.65 %

Net generation 2015[1] / Consumption[1]

		Thermal nuclear	Fossil fuels	– of which lignite	– of which hard coal	– of which gas	– of which oil	– of which mixed fuels	– of which other fossil fuels	Renewable	– of which wind	– of which solar	– of which biomass	– of which other renewable	Hydraulic	– of which renewable	– of which other hydro	Other sources	Total	Pumping	Consumption 2015	Variation (compared with 2014)
		TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	TWh	%
AT	3	–	14.1	–	3.0	7.5	0.9	–	2.7	6.5	3.6	0.4	2.5	–	36.2	32.6	3.7	7.8	**64.6**	5.1	**69.6**	0.4
BA		–	8.5	8.5	–	–	–	–	–	–	–	–	–	–	5.7	5.6	0.0	–	**14.2**	0.0	**12.0**	3.3
BE	5	24.6	26.1	–	4.0	20.8	0.1	–	1.1	13.3	5.4	3.0	2.8	2.0	1.4	0.3	1.1	0.2	**65.5**	1.4	**85.0**	1.5
BG		14.3	21.1	18.8	1.0	1.3	–	–	–	3.0	1.4	1.4	0.2	–	6.2	5.7	0.5	–	**44.5**	0.7	**33.2**	6.5
CH		22.1	2.2	–	–	–	–	–	2.2	2.2	0.1	–	–	2.0	39.6	–	39.6	–	**66.0**	2.3	**63.4**	0.6
CY		–	4.2	–	–	–	4.2	–	–	0.2	0.2	–	–	–	–	–	–	–	**4.4**	–	**4.4**	4.9
CZ		25.3	42.1	32.2	4.8	4.9	0.0	–	0.1	7.1	0.6	2.2	1.9	2.4	3.0	1.8	1.2	–	**77.5**	1.6	**63.4**	2.3
DE	8	86.8	319.5	143.1	107.1	53.2	5.2	10.9	–	150.5	75.7	35.2	38.5	1.2	23.7	16.1	7.6	0.0	**580.4**	8.1	**520.6**	- 1.7
DK		–	10.5	–	6.9	3.5	0.1	–	–	17.0	14.1	0.6	2.3	–	0.0	0.0	–	–	**27.6**	–	**32.4**	- 2.9
EE	9	–	7.6	–	–	–	–	–	7.6	1.5	0.7	–	0.8	–	0.0	0.0	–	–	**9.1**	–	**8.1**	- 0.7
ES		54.8	114.6	4.5	48.6	48.6	13.0	–	–	66.1	48.1	13.3	4.6	0.0	30.8	27.6	3.2	1.3	**267.6**	4.5	**262.9**	1.9
FI		22.3	13.5	–	5.5	5.0	0.2	2.7	–	13.0	2.3	–	10.7	–	16.6	16.6	–	0.8	**66.2**	–	**82.5**	- 1.0
FR		416.8	34.1	–	8.6	22.1	3.4	–	–	36.4	21.1	7.4	7.9	–	58.7	53.9	4.8	–	**546.0**	6.8	**475.4**	2.2
GB	11	65.7	189.9	–	95.0	94.9	–	–	–	59.9	52.4	7.5	–	–	8.0	4.7	3.2	1.4	**324.9**	3.7	**340.2**	1.4
GR		–	26.7	19.4	–	7.3	–	–	–	8.8	3.7	3.6	0.2	1.3	6.1	0.7	5.4	–	**41.6**	0.1	**51.2**	3.9
HR		–	3.2	–	2.1	0.8	–	0.2	–	0.8	0.8	–	–	–	5.7	5.7	–	–	**9.6**	0.2	**17.0**	3.5
HU		14.9	9.2	5.5	0.5	3.2	–	–	–	2.8	0.7	0.0	1.6	0.5	0.2	0.2	–	–	**27.1**	–	**40.8**	3.1
IE		–	18.9	2.5	4.8	11.5	–	–	0.1	6.8	6.5	–	–	0.2	1.1	0.8	0.3	–	**26.8**	0.5	**27.0**	2.9
IS	13	–	0.0	–	–	–	–	–	–	4.7	–	–	–	4.7	13.7	13.7	–	–	**18.3**	–	**18.3**	3.7
IT		–	162.0	–	38.4	91.5	4.2	28.0	–	63.3	14.7	23.9	18.8	5.8	44.6	43.3	1.3	–	**269.8**	1.9	**314.3**	1.9
LT		–	2.3	–	–	1.5	–	0.6	0.3	1.3	0.8	0.1	0.4	–	1.0	0.3	0.7	–	**4.6**	0.9	**10.9**	1.3
LU		–	0.8	–	–	0.8	–	–	–	0.2	0.1	0.1	0.1	–	1.5	0.1	1.4	0.1	**2.7**	1.9	**6.4**	1.8
LV		–	2.6	–	–	2.0	–	0.6	–	0.9	0.1	–	0.4	0.4	1.9	1.9	–	–	**5.4**	–	**7.2**	- 2.2
ME		–	1.4	1.4	–	–	–	–	–	–	–	–	–	–	1.4	–	1.4	–	**2.8**	–	**3.4**	5.1
MK		–	3.3	3.1	–	0.2	–	–	–	0.1	0.1	–	–	–	1.5	–	1.5	–	**4.9**	–	**7.4**	- 5.7
NI	15	–	6.7	–	2.0	4.1	–	0.5	–	2.0	1.8	–	0.1	0.1	–	–	0.0	–	**8.7**	–	**9.4**	0.6
NL		4.0	88.4	–	–	–	–	–	88.4	11.2	7.1	0.1	4.0	–	0.1	0.1	–	–	**103.8**	–	**112.5**	1.4
NO		–	3.5	–	–	3.5	–	–	–	2.5	2.5	–	–	–	139.0	139.0	–	–	**145.0**	2.1	**128.3**	2.5
PL	18	–	132.1	49.5	69.1	4.1	–	–	9.4	17.7	10.5	–	6.7	0.5	2.5	1.8	0.6	–	**152.3**	0.9	**151.1**	2.8
PT		–	23.8	–	13.7	9.8	0.1	–	0.2	14.7	11.3	0.8	2.6	–	9.6	8.5	1.2	–	**48.2**	1.5	**49.0**	0.3
RO		10.7	25.0	14.5	1.7	4.5	–	4.3	–	9.5	7.0	2.0	0.5	–	16.5	16.5	–	–	**61.7**	0.2	**54.8**	2.8
RS		–	30.5	30.5	–	0.0	–	–	–	–	–	–	–	–	10.6	9.9	0.7	–	**41.2**	1.1	**39.3**	2.9
SE		54.3	3.8	–	0.5	1.0	0.1	1.0	1.1	26.4	16.6	–	9.8	–	74.0	74.0	–	–	**158.5**	–	**135.9**	0.3
SI		5.4	3.9	3.8	–	0.0	–	–	0.1	0.5	0.0	0.2	0.2	0.1	4.1	3.8	0.3	0.1	**14.0**	0.4	**13.6**	3.5
SK		14.1	4.6	1.6	0.9	1.8	0.3	–	–	2.2	0.0	0.5	1.1	0.5	4.3	4.0	0.3	0.1	**25.2**	0.4	**27.2**	3.9
ENTSO-E	20	836.0	1360.6	338.9	418.3	409.4	31.9	48.8	113.3	553.1	310.2	102.3	118.7	21.9	569.1	489.2	79.8	11.7	**3330.5**	46.4	**3278.1**	1.2

All footnotes on page 5

Net generating capacity (NGC) as of 31 Dec. 2015[2]

NGC Nuclear	NGC Fossil fuels	– of which lignite	– of which hard coal	– of which gas	– of which oil	– of which mixed fuels	– of which other fossil fuels	NGC Renewable	– of which wind	– of which solar	– of which biomass	– of which other renewable	NGC Hydraulic	– of which renewable	– of which other hydro	NGC Other sources	NGC Total	Representativity of the values		
MW	MW	MW	MW	MW	MW	MW	MW	MW	MW	MW	MW	MW	MW	MW	MW	MW	MW	%		
–	7540	–	1171	4888	334	436	711	2802	1981	404	417	–	13569	13569	–	315	**24226**	100	**AT**	4
–	1578	1578	–	–	–	–	–	–	–	–	–	–	2060	1620	440	–	**3638**	100	**BA**	
5926	7599	–	470	6865	264	–	–	6450	2172	3068	1210	–	1424	116	1308	–	**21399**	100	**BE**	
2000	5706	4199	708	799	–	–	–	1806	701	1041	64	–	3198	2334	864	–	**12710**	100	**BG**	
3333	502	–	–	–	–	–	502	1091	60	756	275	–	13828	12445	1383	245	**18998**	100	**CH**	6
–	1478	1478	–	–	–	–	–	155	155	–	–	–	–	–	–	–	**1633**	100	**CY**	
4040	11140	8334	1200	1606	–	–	–	3194	277	2067	850	–	2253	1081	1172	–	**20627**	100	**CZ**	7
10793	78746	20570	25674	25048	3828	3626	–	89314	43429	38411	6915	559	9610	3270	6340	–	**188463**	100	**DE**	8
0	7540	0	4965	1708	823	44	0	6373	5082	781	510	0	7	7	0	2	**13922**	100	**DK**	
–	2575	–	–	262	85	524	1704	402	301	–	101	–	8	8	–	–	**2984**	100	**EE**	10
7573	46974	1056	9975	32655	3289	–	–	30716	23003	6967	747	–	20336	17036	3300	588	**106187**	100	**ES**	
2752	8420	–	3295	1743	1695	1687	–	3001	1082	–	1919	–	3263	3263	–	245	**17681**	100	**FI**	
63130	22553	–	3007	10901	8645	–	–	18206	10312	6191	1703	–	25421	23442	1979	–	**129310**	100	**FR**	
8981	47914	–	17926	29050	938	–	–	18880	8503	9000	1377	–	3866	1122	2744	–	**79641**	80	**GB**	11
–	9922	4456	–	4768	698	–	–	4271	1775	2444	52	–	3242	224	3018	135	**17570**	100	**GR**	
–	1770	–	325	496	320	629	–	496	384	44	26	42	2112	2112	–	–	**4378**	100	**HR**	
1887	5541	1007	–	4124	410	–	–	603	328	29	246	–	57	57	–	88	**8176**	100	**HU**	12
–	6409	346	855	4270	926	12	–	2474	2400	–	7	67	534	242	292	251	**9668**	100	**IE**	
–	11	–	–	–	11	–	–	663	2	–	–	661	1972	1972	–	–	**2635**	100	**IS**	14
–	69700	0	8800	51500	6000	3400	–	32330	8750	19100	3700	780	22150	–	22150	–	**124180**	100	**IT**	
–	2332	–	–	553	–	1779	–	436	290	69	77	–	126	126	–	900	**3794**	100	**LT**	
–	495	–	–	495	–	–	–	187	60	116	11	–	1334	44	1290	21	**2037**	100	**LU**	
–	1144	–	–	1031	–	113	–	183	70	–	113	–	1556	1556	–	–	**2883**	100	**LV**	
–	220	220	–	–	–	–	–	–	–	–	–	–	660	10	650	–	**880**	100	**ME**	
–	1157	718	–	250	189	–	–	36	36	–	–	–	539	–	539	–	**1732**	100	**MK**	
–	5904	–	3836	2064	4	–	–	1550	1447	–	43	60	12	12	–	–	**7466**	100	**NI**	
486	25572	–	5658	19914	–	–	–	5468	3641	1429	398	–	38	38	–	674	**32238**	100	**NL**	16
–	1600	–	–	1600	–	–	–	892	860	–	32	–	31200	31200	–	–	**33692**	100	**NO**	17
–	29232	8525	17016	1354	–	–	2337	6087	5186	87	612	202	2355	942	1413	–	**37674**	100	**PL**	19
–	6519	–	1756	4698	47	–	18	5868	4826	429	613	–	6146	6146	–	–	**18533**	100	**PT**	
1300	8496	3777	1148	1850	–	1721	–	4285	2923	1249	112	–	6339	6339	–	–	**20420**	100	**RO**	
–	5549	5238	–	311	–	–	–	–	–	–	–	–	3009	2389	620	–	**8558**	100	**RS**	
9714	4501	–	225	879	3277	120	–	9552	6029	104	2978	441	16184	16184	–	–	**39951**	100	**SE**	
696	1470	921	–	549	–	–	–	284	3	263	17	–	1233	1053	180	40	**3722**	100	**SI**	
1940	2718	568	440	1093	195	422	–	904	3	532	259	110	2533	1617	916	–	**8095**	100	**SK**	
124551	440526	62991	108450	217324	31978	14513	5271	258958	136071	94582	25383	2922	202173	151575	50598	3503	**1029700**		**ENTSO-E**	20

1 All values are calculated to represent 100 % of the national values. When 0.0 is displayed, the value is higher than 0.00 but lower than 0.05 TWh.

2 NGC from 2014 for CH and NI and from 2013 for MK.

3 Renewable hydro: Production run-of-river plus production of storage and pumped storage minus 70 % of pumped energy.

4 711 MW of Fossil Fuels NGC not attributable. All hydro power plants are considered as renewable as also pump storage plants have natural inflow.

5 The reported figures are best estimates based on actual measurements and extrapolations.

6 Hydraulic pumps.

7 Gas including Coal-Derived Gas. Biomass including biogas.

8 RES (other than hydro) from TSO data, rest from official statistics.

9 Of which other fossil fuel = oil shale.

10 NGC Fossil Fuels includes also oil shale which is 1704 MW.

11 All data with the country code GB represents monthly statistical data as sum of England, Scotland and Wales. Values from December 2015 replaced by values from 2014.

12 Other renewable and waste.

13 Diesel production (backup energy in disturbance).

14 Geothermal 661 MW.

15 All data with the country code NI represents the monthly statistical data of GB Northern Ireland.

16 Other Sources represents Waste.

17 Fossil fuels NGC, of which 1150 MW unavailable.

18 The coverage ratio used for recalculating energy in each subcategory amounts to 99 % and based on historical differences of total generation between data available to PSE and official statistics. Biomass generation includes biomass co-fired in conventional thermal units. Other renewable represents biogas and waste. Other fossil fuel represents industry.

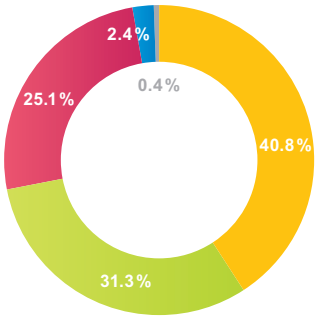
19 Operational data available to TSO. Other fossil fuel represents industry (different fossil fuels). Other renewable represent biogas and waste.

20 Calculated sum of the ENTSO-E member TSOs' countries

Generation

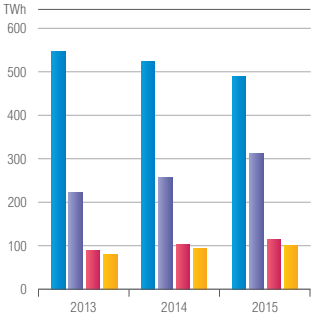
Generation mix in ENTSO-E member TSOs' countries in 2015[1]

	TWh
Fossil fuels net generation (lignite and hard coal, gas, oil, mixed fuels, peat)	1360.60
Renewable net generation (hydro, wind, solar, biomass, geothermal)	1042.45
Thermal nuclear net generation	836.04
Hydraulic net generation (except renewable part)	79.82
Non-identifiable net generation	11.69



ENTSO-E renewable generation[1]

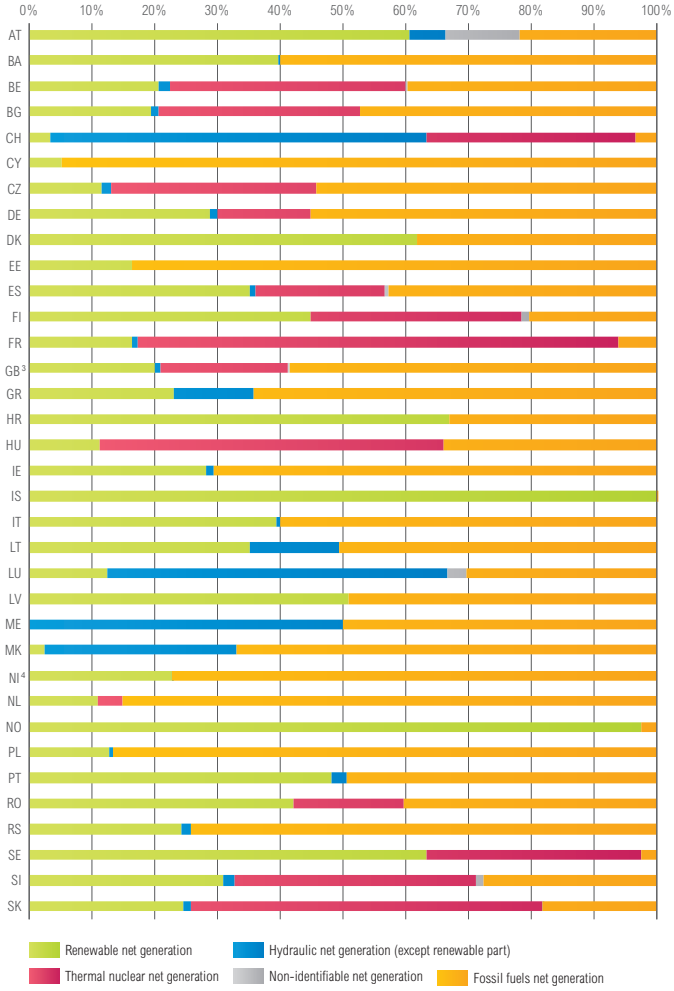
	year	TWh
Renewable net generation	2013	962.31
	2014	997.16
	2015	1042.45
of which hydro	2013	545.83
	2014	522.48
	2015	489.25
of which wind	2013	222.81
	2014	254.71
	2015	310.59
of which biomass	2013	87.97
	2014	103.65
	2015	116.18
of which solar	2013	80.95
	2014	95.91
	2015	101.96



ENTSO-E fossil fuels generation[1]

	year	TWh
Fossil fuels net generation	2013	1437.83
	2014	1332.26
	2015	1360.60
of which hard coal	2013	441.08
	2014	437.00
	2015	418.28
of which gas	2013	386.16
	2014	371.68
	2015	409.41
of which lignite	2013	329.65
	2014	338.53
	2015	338.92
of which oil	2013	42.78
	2014	40.45
	2015	31.88
of which mixed fuels	2013	45.45
	2014	37.84
	2015	48.80

Share of energy produced of each member TSOs' country 2015 in %[2]



[1] All values are calculated to represent 100 % of the national values.

[2] Share of energy produced, based on the net generation for each TSO as a member of ENTSO-E per the tables "ENTSO-E in figures" on page 2–5.

[3] All data with the country code GB represents monthly statistical data as sum of England, Scotland and Wales.

[4] All data with the country code NI represents the monthly statistical data of GB Northern Ireland.

Physical energy flows

	Sum of imports[1]	Sum of exports[1]	Balance (imp−exp)
AT	29 368	19 311	10 057
BA	3 874	6 005	-2 131
BE	23 737	2 718	21 019
BG	4 238	14 704	-10 466
CH	33 620	33 804	-184
CY[2]	–	–	–
CZ	16 139	28 661	-12 522
DE	33 563	85 351	-51 788
DK	15 898	9 642	6 256
EE	5 349	6 273	-924
ES	14 956	15 089	-133
FI	21 200	5 426	15 774
FR	9 917	72 904	-62 987
GB[3]	23 507	2 436	21 071
GR	11 091	1 473	9 618
HR	13 167	5 529	7 638
HU	19 936	6 249	13 687
IE	1 776	1 050	726
IS[2]	–	–	–
IT	50 826	3 423	47 403
LT	7 938	730	7 208
LU	7 524	1 917	5 607
LV	5 243	3 424	1 819
ME	3 092	2 500	592
MK	5 288	2 777	2 511
NI[4]	1 296	708	588
NL	30 883	21 775	9 108
NO	7 208	21 761	-14 553
PL	14 459	14 793	-334
PT	8 077	5 811	2 266
RO	1 433	8 158	-6 725
RS	6 615	7 365	-750
SE	12 514	34 302	-21 788
SI	9 039	8 976	63
SK	14 966	12 612	2 354
ENTSO-E	**467 740**	**467 649**	**91**

Physical flow values in GWh

[1] Consolidated yearly values might differ from detailed flow data from the ENTSO-E database due to ex-post consolidation taking into account national statistical resources.

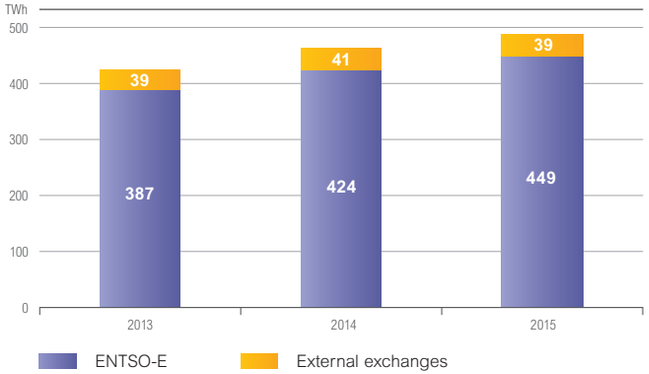
[2] Isolated system; therefore no exchange data.

[3] All data with the country code GB represents monthly statistical data as sum of England, Scotland and Wales.

[4] All data with the country code NI represents the monthly statistical data of GB Northern Ireland.

Development of exchanges

Development of overall cross-border exchanges of ENTSO-E member TSOs' countries during the past 3 years



Overview electricity exchanges for the past 3 years in GWh

	All Exchanges	ENTSO-E	External[1]
2013	426148	387251	38897
2014	464395	423650	40745
2015	488227	449036	39191

[1] External exchanges include Albania, Belarus, Moldavia, Morocco, Russia, Turkey, Ukraine and Ukraine-West.

Publisher: Secretariat of ENTSO-E AISBL
Avenue de Cortenbergh 100, 1000 Brussels – Belgium

Contribution: ENTSO-E Statistical Data Correspondents and Data Expert Group members

Design: DreiDreizehn Werbeagentur GmbH, Berlin, www.313.de

Printed by: Kehrberg Druck Produktion Service, Berlin

May not be reproduced without prior permission from ENTSO-E

Highest and lowest hourly load values of each country 2015[1]

	Highest load Date	Time[2]	value (in MW)	Lowest load Date	Time[2]	value (in MW)
AT	24.11	17:00	11386	23.08	03:00	4722
BA	31.12	17:00	2105	02.05	03:00	858
BE[3]	22.01	18:00	13129	02.08	06:00	6556
BG	08.01	18:00	7100	07.06	05:00	2781
CH	12.02	08:00	10155	02.08	06:00	4875
CY	04.08	14:00	954	01.06	05:00	270
CZ	09.02	12:00	9982	02.08	05:00	4428
DE	24.11	17:00	79893	25.05	03:00	36146
DK	10.06	10:00	5844	26.04	03:00	1915
EE	07.01	15:00	1394	27.07	04:00	498
ES	04.02	20:00	40324	25.12	05:00	18041
FI	22.01	07:00	13584	19.09	02:00	5194
FR	06.02	18:00	91611	16.08	06:00	29299
GB[4]	19.01	18:00	59576	12.07	06:00	20871
GR	30.07	12:00	9813	25.10	14:00	3599
HR	22.07	12:00	2950	22.06	05:00	1160
HU	08.07	12:00	6106	25.05	05:00	2857
IE	14.01	17:00	4662	19.07	07:00	1804
IS	25.02	13:00	2327	08.08	06:00	1622
IT	22.07	15:00	59648	26.12	04:00	18785
LT	07.01	16:00	1748	12.07	04:00	784
LU	07.07	11:00	1087	02.08	03:00	339
LV	08.01	15:00	1225	25.06	03:00	438
ME	22.07	13:00	583	11.05	04:00	225
MK	08.01	23:00	1439	02.06	03:00	530
NI[5]	14.01	18:00	1737	14.07	06:00	538
NL	06.01	17:00	17761	02.08	06:00	8892
NO	04.02	08:00	22530	23.08	05:00	9527
PL	07.01	17:00	23069	26.12	00:00	10143
PT	07.01	20:00	8618	05.04	07:00	3419
RO	08.01	18:00	8488	12.04	14:00	3799
RS	31.12	17:00	6879	10.05	05:00	2486
SE	05.02	17:00	23395	19.07	05:00	9242
SI	05.02	11:00	2086	01.05	03:00	960
SK	25.11	17:00	4145	02.08	05:00	2231
ENTSO-E[6]	**05.02**	**18:00**	**528093**	**16.08**	**06:00**	**240673**

[1] Hourly load values represent the average of the hour, they are calculated to represent 100% of the national values.

[2] It is the hour following the given time (00:00 meaning the hour from 00:00 to 01:00)

[3] The reported figures are best estimates based on actual measurements and extrapolations.

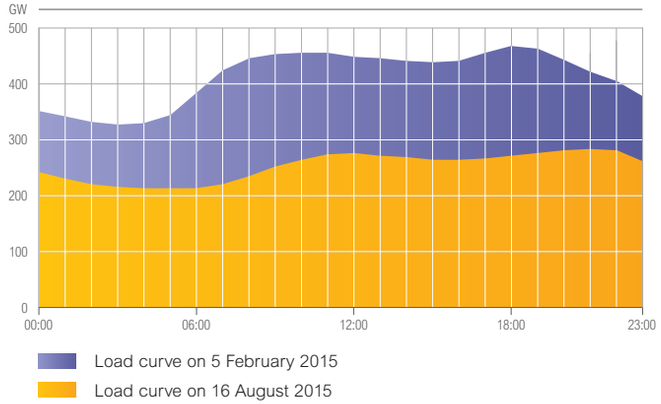
[4] All data with the country code GB represents monthly statistical data as sum of England, Scotland and Wales.

[5] All data with the country code NI represents the monthly statistical data of GB Northern Ireland.

[6] Calculated as sum of the ENTSO-E member TSOs' monthly hourly load values.

ENTSO-E peak load 2015

ENTSO-E load diagram on the days of the highest and lowest load values[1,2]



Country values in MW on the days of highest and lowest ENTSO-E load values[1,2]

Time[3]	05.02.15 at 18:00	16.08.15 at 06:00		05.02.15 at 18:00	16.08.15 at 06:00		05.02.15 at 18:00	16.08.15 at 06:00
AT	10977	5377	**FR**	90970	29299	**MK**	1151	600
BA	1877	996	**GB**[4]	57988	23043	**NI**[5]	1543	571
BE	13091	6856	**GR**	7358	4561	**NL**	17272	9430
BG	5899	3119	**HR**	2714	1560	**NO**	21433	9596
CH	9658	5256	**HU**	5796	3566	**PL**	22245	11299
CY	604	511	**IE**	4470	1927	**PT**	7394	4000
CZ	9828	4788	**IS**	2303	1913	**RO**	8023	4362
DE	79455	38978	**IT**	50019	20614	**RS**	6303	2907
DK	5474	2606	**LT**	1600	889	**SE**	22971	9805
EE	1267	591	**LU**	812	383	**SI**	2085	1090
ES	37650	20051	**LV**	1107	560	**SK**	4038	2428
FI	12200	6817	**ME**	518	324	*****	**528 093**	**240 673**

***ENTSO-E**

[1] Calculated load values as sum of the ENTSO-E member TSOs' countries.

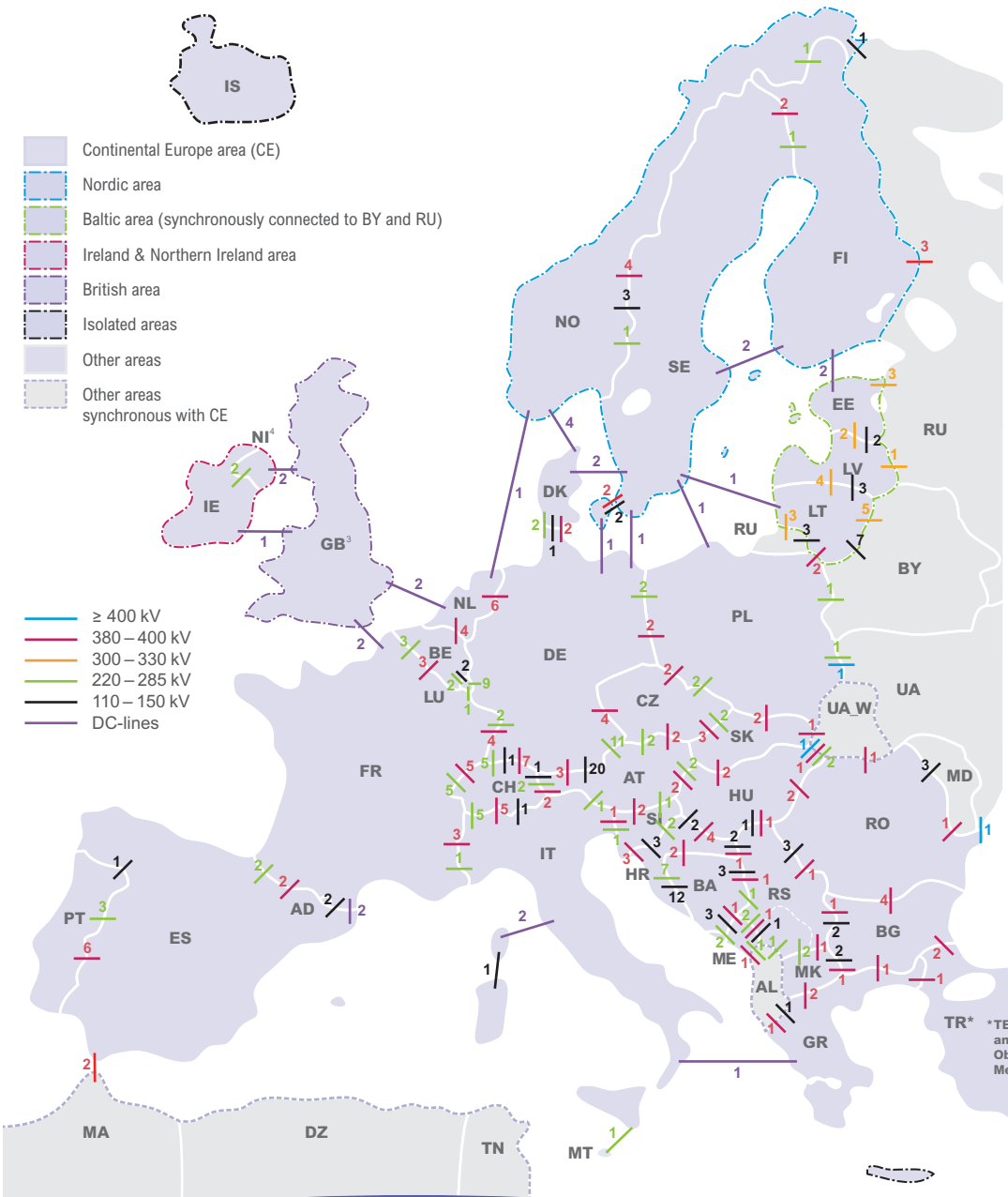
[2] All values are calculated to represent 100% of the national values.

[3] It is the hour following the given time (00:00 meaning the hour from 00:00 to 01:00)

[4] All data with the country code GB represents monthly statistical data as sum of England, Scotland and Wales.

[5] All data with the country code NI represents the monthly statistical data of GB Northern Ireland.

Grid information



Number of circuits on cross-frontier transmission lines as of 31 December 2015 in the ENTSO-E area[1,2]

ENTSO-E Overview circuit length in km

	Length of AC circuits	of which AC cables	Sum of DC cables
Over 400 kV	4697	1749	2334
350 – 400 kV	152486	1432	1047
300 – 330 kV	10962	225	62
220 – 285 kV	140407	3610	2233
110 – 160 kV	–	–	105
Sum	**308552**	**7016**	**5781**

Number of cross frontier lines in the ENTSO-E area

	AC	DC
Over 400 kV	7	8
380 – 400 kV	121	5
300 – 330 kV	18	3
200 – 285 kV	93	10
110 – 150 kV	88	1
Under 110 kV	1	–
Sum	**328**	**27**

[1] Non geographic loacation of lines.

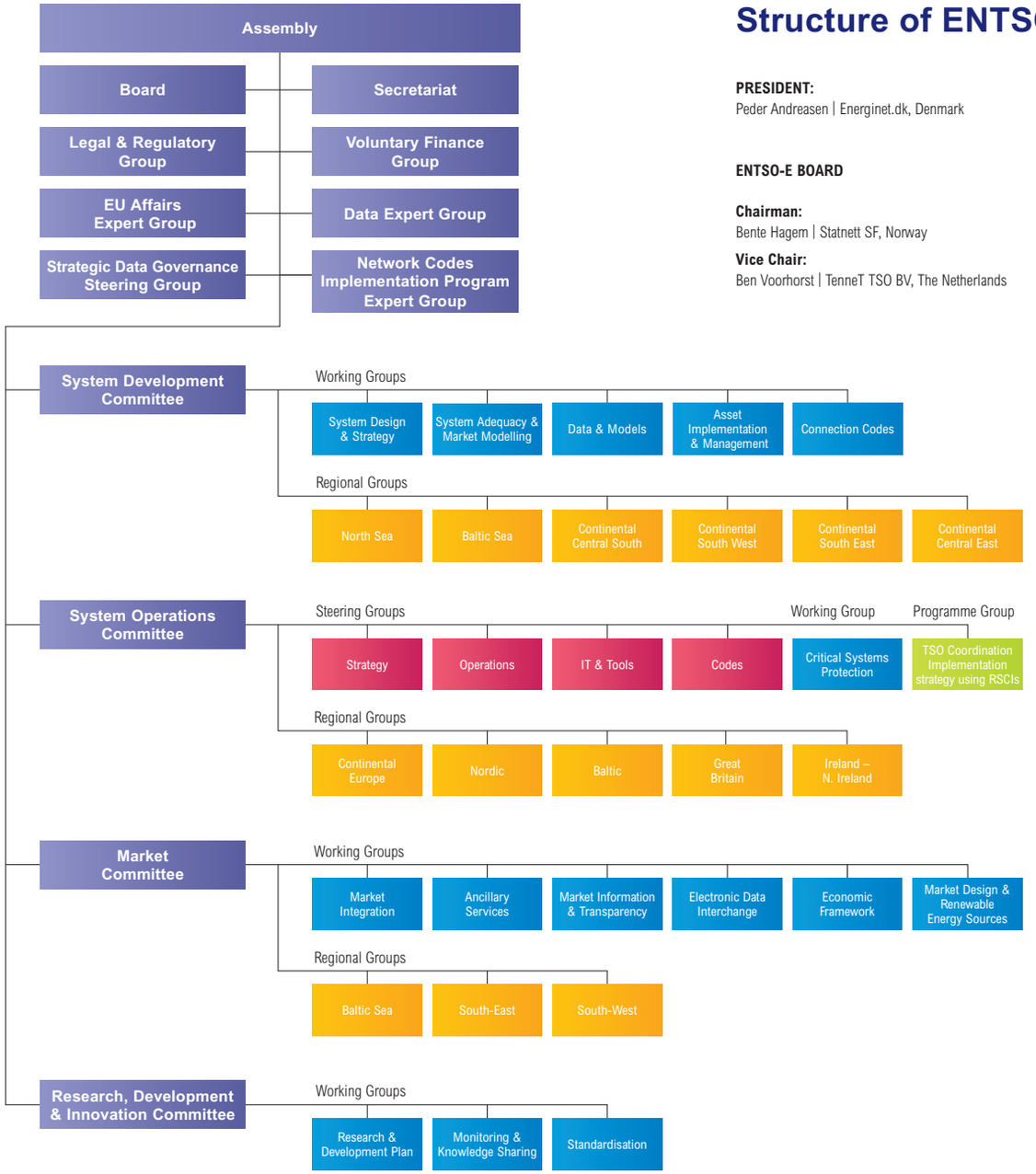
[2] Only lines operated by TSOs are taken into account.

[3] All data with the country code GB represents statistical data as sum of England, Scotland and Wales.

[4] All data with the country code NI represents statistical data GB Northern Ireland.

Members of ENTSO-E

AT	Austria	APG VÜN	Austrian Power Grid AG Vorarlberger Übertragungsnetz GmbH
BA	Bosnia and Herzegovina	NOS BiH	Nezavisni operator sustava u Bosni i Hercegovini
BE	Belgium	Elia	Elia System Operator NV/SA
BG	Bulgaria	ESO	Electroenergien Sistemen Operator EAD
CH	Switzerland	Swissgrid	Swissgrid AG
CY	Cyprus	Cyprus TSO	Cyprus Transmission System Operator
CZ	Czech Republic	ČEPS	ČEPS, a.s.
DE	Germany	TransnetBW TenneT DE Amprion 50Hertz	TransnetBW GmbH TenneT TSO GmbH Amprion GmbH 50Hertz Transmission GmbH
DK	Denmark	Energinet.dk	Energinet.dk
EE	Estonia	Elering AS	Elering AS
ES	Spain	REE	Red Eléctrica de España S.A.U.
FI	Finland	Fingrid	Fingrid Oyj
FR	France	RTE	Réseau de transport d´électricité
GB	United Kingdom	National Grid SONI (NI) SHE Transmission SPTransmission	National Grid Electricity Transmission plc System Operator for Northern Ireland Ltd Scottish Hydro Electric Transmission Ltd Scottish Power Transmission plc
GR	Greece	IPTO	Independent Power Transmission Operator S.A.
HR	Croatia	HOPS	HOPS d.o.o.
HU	Hungary	MAVIR ZRt.	MAVIR Magyar Villamosenergia-ipari Átviteli Rendszerirányító Zártkörűen Működő Részvénytársaság
IE	Ireland	EirGrid	EirGrid plc
IS	Iceland	Landsnet	Landsnet hf
IT	Italy	Terna	Terna – Rete Elettrica Nazionale SpA
LT	Lithuania	Litgrid	Litgrid AB
LU	Luxembourg	Creos Luxembourg	Creos Luxembourg S.A.
LV	Latvia	Augstsprieguma tīkls	AS Augstsprieguma tīkls
ME	Montenegro	CGES AD	Crnogorski elektroprenosni sistem AD
MK	FYROM	MEPSO	Macedonian Transmission System Operator AD
NL	The Netherlands	TenneT NL	TenneT TSO B.V.
NO	Norway	Statnett	Statnett SF
PL	Poland	PSE	PSE S.A.
PT	Portugal	REN	Rede Eléctrica Nacional, S.A.
RO	Romania	Transelectrica	C.N. Transelectrica S.A.
RS	Serbia	EMS	JP Elektromreža Srbije
SE	Sweden	Svenska Kraftnät	Svenska Kraftnät
SI	Slovenia	ELES	Elektro Slovenija d.o.o.
SK	Slovak Republic	SEPS	Slovenská elektrizačná prenosová sústava, a.s.
Observer member			
TR	Turkey	TEIAŞ	Türkiye Elektrik İletim A.Ş.



Structure of ENTSO-E

PRESIDENT:
Peder Andreasen | Energinet.dk, Denmark

VICE-PRESIDENT:
Matteo del Fante | Terna, Italy

ENTSO-E BOARD

Chairman:
Bente Hagem | Statnett SF, Norway

Vice Chair:
Ben Voorhorst | TenneT TSO BV, The Netherlands

Zbyněk Boldiš | ČEPS, Czech Republic
Kamilla Csomai | MAVIR, Hungary
Klaus Kleinekorte | Amprion, Germany
Matt Golding | National Grid, United Kingdom
Thomas Karall | APG, Austria
Santiago Marin | REE, Spain
Jukka Ruusunen | Fingrid, Finland
Tomasz Sikorski | PSE, Poland
Thomas Tillwicks | Swissgrid, Switzerland
Jean Verseille | RTE, France

ENTSO-E at a Glance

ENTSO-E, the European Network of Transmission System Operators, represents 42 electricity transmission system operators (TSOs) from 35 countries across Europe. ENTSO-E was established and given legal mandates by the EU legislation.

ENTSO-E promotes closer cooperation across Europe's TSOs to support the implementation of EU energy policy and achieve Europe's energy & climate policy objectives, which are changing the very nature of the power system.

Reliable. Sustainable. Connected.

ENTSO-E contributes to the achievement of these objectives mainly through:

- the drafting and implementation of network codes;
- the development of pan-European network plans (TYNDPs);
- the technical cooperation between TSOs and between TSOs and DSOs;
- the publication of outlook reports for electricity generation;
- the publication of fundamental data on the EU electricity markets;
- the coordination of R&D plans.

ENTSO-E is the focal point for all technical, market and policy issues relating to TSOs and the European network, interfacing with power system users, EU institutions, regulators and national governments. **Through its work, ENTSO-E is helping to build the world's largest electricity market, the benefits of which will not only be felt by all those in the energy sector but also by Europe's overall economy, today and into the future.**

Visit www.entsoe.eu for more EU electricity statistics, data, facts & figures and our latest position papers, publications and press releases. By visiting our website, subscribing to our news alert or following us on twitter you will be able to keep up-to-date with the development of our key work products which all make the European energy policy progress to the benefit of consumers.

Contact

ENTSO-E AISBL
Avenue de Cortenbergh 100 · 1000 Brussels · Belgium

Tel +32 2 741 09 50 · Fax +32 2 741 09 51
info@entsoe.eu · www.entsoe.eu

entsoe
Reliable Sustainable Connected